Student Record System · Software Engineering Mini Project

Objective : Covered all 25 functional and 7 non-functional requirements in the SRS with the existing GitHub issues to identify coverage gaps and required updates.

Functional requirements (SRS-F-001 to SRS-F-025)

Requirement	GitHub issue(s)	Coverage status
F-001 — Create student records	#1, #15	Covered
F-002 — View student information	#2	Covered
F-003 — Update student records	#3	Covered
F-004 — Prevent duplicate student IDs	#15, #12	Covered
F-005 — Maintain subjects	#7	Covered
F-006 — View subjects	#7	Covered
F-007 — Assign subjects to students	#7	Covered
F-008 — Enter grades	#7	Covered
F-009 — Validate grades	#7, #12	Covered
F-010 — Update grades	#7	Covered
F-011 — Calculate GPA and CGPA	#8	Covered
F-012 — Record attendance	#9	Covered
F-013 — Update attendance	#9	Covered

F-014 — Calculate attendance percentage	#9	Covered
F-015 — Search by student ID or name	#2	Covered
F-016 — Generate academic reports	#11	Covered
F-017 — View grade sheets and attendance summaries	#11	Covered
F-018 — Authenticate users	#5	Covered
F-019 — Role-based access control	#6	Covered
F-020 — Prevent students modifying protected records	#6	Covered
F-021 — Persist student and academic data	#10	Covered
F-022 — Validate required information	#3, #12	Covered
F-023 — Maintain data consistency	#10, #12	Partially covered (Reason: Consistency rules for related records during updates and deletions still need confirmation and verification.)
F-024 — Logout and exit	New logout issue	Covered
F-025 — Manage authorized user accounts	#5, #6	Partially covered

Non-functional requirements (SRS-NF-001 to SRS-NF-007)

Requirement	**GitHub issue(s)**	**Coverage status**

NF-001 — Response time within 3 seconds for 95% of requests	New performance issue	Covered
NF-002 — Reliability and prevention of data loss	#10	Partially covered (Reason: Specific reliability, failure-handling, and data-recovery criteria and verification tests still need to be defined.)
NF-003 — Protect stored personal information	#13	Covered
NF-004 — Audit logs retained for five years	#13	Covered
NF-005 — Accessibility and WCAG 2.1 AA	New accessibility issue	Covered
NF-006 — Performance at expected record volumes	#10, new performance issue	Covered
NF-007 — Distinct permissions for each role	#6	Covered

Note: “Covered” here means the requirement is explicitly documented in an issue description or acceptance criteria. It does not mean the feature has been implemented, tested, or proven to work.

Gaps Identified and Actions Taken:

The existing GitHub backlog was reviewed against the functional and non-functional requirements specified in the Software Requirements Specification (SRS) for the Student Record System. The review identified requirements that needed clearer descriptions, acceptance criteria, and verification steps.

The following actions were taken:

- **Existing issue descriptions updated:** Descriptions and acceptance criteria were added to Issues #1–#13 to clarify the expected functionality, validation rules, security controls, data storage, and testing requirements.

- **Student record management:** Requirements for student creation, viewing, searching, updating, and deletion were documented more clearly.
- **Academic functionality:** Requirements for subject management, grade entry and updates, GPA/CGPA calculation, attendance tracking, and academic reports were specified.
- **Authentication and security:** Requirements for login, role-based access control, secure password storage, session management, and audit logging were documented.
- **Additional requirements identified:** New issues were created to address secure logout and session termination, accessibility support, and performance requirements.
- **Data validation and integrity:** Acceptance criteria were added for required fields, duplicate student IDs, invalid grades, and consistency between related records.

Some areas require further verification or clarification. In particular, administrator management of user accounts, consistency of related academic records during updates or deletion, reliability and recovery from data loss, and the verification of security and performance requirements should be reviewed to ensure complete coverage of the SRS.

The resulting backlog provides clearer descriptions of the expected behaviour and criteria for verifying the requirements. The coverage matrix records the relationship between SRS requirement IDs and their corresponding GitHub issues.

➔ **F-025 — Administrator user-account management:** Role-based access control (GitHub Issue #6) defines what each user role is permitted to do. F-025 separately requires administrators to create, modify, and deactivate authorized user accounts. Issue #6 has been flagged for clarification so that these account-management actions have explicit acceptance criteria and verification tests. This requirement remains subject to confirmation.

➔ The existing GitHub backlog was reviewed against the SRS requirements. Issue descriptions and acceptance criteria were clarified where needed, and additional issues were created for logout, accessibility, and performance. Issue #14, the testing backlog item, was also identified for completeness review. These changes improve requirements traceability; they do not establish that all functionality has been implemented or tested.

Conclusion:

The requirements review compared the Student Record System's functional and non-functional requirements with the existing GitHub issue backlog. The review helped identify areas where issue descriptions and acceptance criteria needed improvement, as well as requirements that required additional issues or further clarification.

The updates made to the GitHub issues improve the traceability between the SRS and the development backlog and provide a clearer basis for implementation and testing. The coverage matrix can be used by the project team to track requirements and identify outstanding work.

However, documenting a requirement in a GitHub issue does not establish that the corresponding functionality has been implemented or tested successfully. Implementation, verification, and validation must be completed separately to confirm that the system satisfies the specified requirements.

Overall, this review establishes a more structured basis for requirements tracking and helps the team identify remaining gaps before considering the Student Record System complete.

References and Evidence

1. Software Requirements Specification (SRS): Student Record System project SRS, containing functional requirements SRS-F-001 to SRS-F-025 and non-functional requirements SRS-NF-001 to SRS-NF-007.

2. GitHub Repository: [Student Record System](https://github.com/Student-Record-System-project/Student-Record-System)

3. GitHub Project Board:** [Student Record System — Sprint Planning](https://github.com/orgs/Student-Record-System-project/projects/2/views/1)

The GitHub issues and project board were used to document requirement coverage, acceptance criteria, dependencies, and outstanding work. The issue descriptions represent planned requirements and verification criteria; they are not evidence that implementation or testing has been completed.

Additional review findings

During the final review, two requirements were identified for further clarification:

- **SRS-F-025 — User account management:** Issue #6 was flagged to ensure that administrators can create, modify, and deactivate authorized user accounts. The

team was asked to include explicit acceptance criteria and test cases for these operations.

- **SRS-F-023 — Data consistency:** Issue #10 was flagged to ensure that updates and deletions of student or subject records do not leave inconsistent or orphaned grade and attendance records. The team was asked to verify the relevant update and deletion rules through test cases.

These clarifications were added as comments to the existing GitHub issues to preserve the current issue descriptions and avoid creating duplicate work. Both items remain subject to confirmation by the project team.

Additional Issues Created

- **Issue #16 — Implement accessibility support:** Addresses SRS-NF-005 accessibility requirements.
- **Issue #17 — Implement performance requirements and response-time testing:** Addresses SRS-NF-001 and SRS-NF-006 performance requirements.
- **Issue #15 — Include required student fields and duplicate-ID rejection:** Supports SRS-F-001 and SRS-F-004.

Existing testing backlog: Issue #14 — Test functional and non-functional requirements. This issue should also be reviewed to ensure that the planned tests cover the functional and non-functional requirements and their verification criteria.

Issues:

GitHub issue	Requirement mapping
#15 — Include required student fields and duplicate-ID rejection	F-001, F-004
#16 — Implement accessibility support	NF-005
#17 — Implement performance requirements and response-time testing	NF-001, NF-006
#18 — Implement secure logout and session termination	F-024
#14 — Test functional and non-functional requirements	Testing backlog; review for completeness